

SOSNOVO-OZERSKOYE, Russian Federation

WMO: 307450

Lat: 52.538N

Lon: 111.558E

Elev: 951

StdP: 90.41

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.5	-34.0	-38.7	0.1	-36.2	-36.3	0.1	-33.8	11.3	-18.3	10.0	-18.2	1.3	180	0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	27.0	17.1	25.2	16.5	23.4	15.6	18.3	24.5	17.4	23.1	16.5	21.8	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.9	20.8	15.2	12.1	19.9	14.3	11.4	19.3	56.0	24.4	52.8	23.0	50.0	22.0	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.5	8.3	-39.3	30.5	3.3	2.4	-41.7	32.2	-43.6	33.6	-45.5	34.9	-47.9	36.7		
(5)			-39.4	19.9	3.3	1.1	-41.7	20.7	-43.7	21.3	-45.5	22.0	-47.9	22.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.5	-23.2	-19.1	-11.0	-1.1	5.9	14.4	17.2	14.5	7.2	-1.8	-13.0	-20.8	(6)
(7)		DBStd	14.84	5.68	5.86	6.34	4.59	4.10	3.67	2.75	3.08	3.95	4.71	6.42	5.99	(7)
(8)		HDD10.0	5093	1030	814	651	333	139	8	0	4	103	367	691	954	(8)
(9)		HDD18.3	7628	1288	1047	909	583	385	125	54	123	336	625	941	1213	(9)
(10)		CDD10.0	537	0	0	0	0	13	139	224	143	17	0	0	0	(10)
(11)		CDD18.3	30	0	0	0	0	0	6	20	4	0	0	0	0	(11)
(12)		CDH23.3	408	0	0	0	0	5	109	226	66	2	0	0	0	(12)
(13)		CDH26.7	67	0	0	0	0	1	17	42	8	0	0	0	0	(13)
(14)	Wind	WSAvg	3.4	2.9	3.2	3.4	4.1	3.9	3.4	3.1	3.0	3.8	3.7	3.5	2.9	(14)
(15)	Precipitation	PrecAvg	329	2	2	4	14	22	53	96	79	36	9	6	4	(15)
(16)		PrecMax	494	8	7	15	55	83	168	181	166	85	28	23	10	(16)
(17)		PrecMin	200	0	0	0	0	1	14	29	18	8	0	0	0	(17)
(18)		PrecStd	86	2	2	4	13	19	33	46	30	20	8	5	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.9	-2.7	7.6	15.9	23.4	28.8	30.0	27.5	22.8	13.4	3.5	-5.4	(19)
(20)		0.4%	MCWB	-8.7	-5.2	2.0	6.8	12.2	16.8	17.9	17.5	13.6	7.4	0.0	-6.6	(20)
(21)		2%	DB	-11.0	-5.8	3.6	11.7	19.9	26.1	27.7	25.1	19.3	10.2	0.6	-8.2	(21)
(22)		2%	MCWB	-11.7	-7.5	-0.2	4.8	10.2	16.2	17.6	16.4	12.2	5.0	-1.9	-9.2	(22)
(23)		5%	DB	-13.3	-8.0	1.1	8.8	16.9	24.3	25.8	23.0	16.7	7.5	-1.6	-10.0	(23)
(24)		5%	MCWB	-14.0	-9.3	-2.0	3.1	8.8	15.4	17.1	15.5	10.9	3.3	-3.5	-10.8	(24)
(25)		10%	DB	-15.4	-10.1	-1.1	6.4	14.3	22.3	24.0	21.1	14.4	5.1	-3.8	-12.1	(25)
(26)		10%	MCWB	-16.0	-11.1	-3.6	1.7	7.4	14.5	16.5	14.9	9.5	1.7	-5.3	-12.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.6	-5.0	2.2	7.4	13.1	18.8	19.9	18.7	14.6	7.7	0.1	-6.6	(27)
(28)		0.4%	MCDWB	-7.9	-2.7	6.7	14.8	22.0	26.4	26.8	24.1	20.1	13.2	3.1	-5.5	(28)
(29)		2%	WB	-11.7	-7.4	-0.1	5.1	10.9	17.1	18.7	17.6	12.8	5.3	-1.8	-9.3	(29)
(30)		2%	MCDWB	-11.0	-5.8	3.5	11.3	18.7	24.0	25.5	22.8	18.1	9.8	0.5	-8.3	(30)
(31)		5%	WB	-14.0	-9.3	-1.9	3.3	9.2	16.0	17.9	16.5	11.4	3.5	-3.4	-10.8	(31)
(32)		5%	MCDWB	-13.3	-8.1	1.0	8.2	15.9	22.5	23.8	21.3	15.9	7.1	-1.6	-10.1	(32)
(33)		10%	WB	-15.9	-11.1	-3.7	1.9	7.7	15.0	17.0	15.5	9.9	1.9	-5.3	-12.8	(33)
(34)		10%	MCDWB	-15.4	-10.1	-1.1	6.0	13.9	21.0	22.5	20.1	14.2	4.8	-3.9	-12.2	(34)
(35)	Mean Daily Temperature Range		MDBR	9.0	11.5	12.9	10.9	11.4	11.5	11.1	10.7	10.8	9.6	8.7	8.4	(35)
(36)		5% DB	MCDBR	9.0	12.0	13.1	15.1	17.3	16.1	15.1	15.2	15.7	13.5	9.6	7.5	(36)
(37)			MCWBR	8.3	10.6	9.8	9.5	9.6	7.9	7.1	7.5	9.1	8.9	7.7	7.0	(37)
(38)			MCDBR	9.0	11.9	12.7	14.0	15.8	13.9	12.2	12.5	14.3	12.6	9.3	7.8	(38)
(39)		5% WB	MCWBR	8.4	10.5	9.6	9.1	9.1	7.2	6.3	6.7	8.4	8.6	7.6	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.240	0.266	0.300	0.362	0.459	0.445	0.395	0.364	0.325	0.287	0.261	0.241	(40)	
(41)		taud	2.194	2.170	2.164	2.187	1.959	2.115	2.333	2.401	2.456	2.465	2.289	2.162	(41)	
(42)		Ebn at Noon	764	848	886	868	792	810	850	860	860	829	732	682	(42)	
(43)		Edh at Noon	65	92	115	129	172	148	118	104	87	69	60	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.26	2.31	3.81	5.14	5.72	6.01	5.56	4.73	3.65	2.34	1.33	0.94	(44)	
(45)		RadStd	0.06	0.08	0.10	0.28	0.33	0.38	0.46	0.26	0.26	0.20	0.08	0.05	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-2.12	-2.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	-1.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page